



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Spring 2026

CSE 4611/CSI 411: Compiler Design/Compiler

Total Marks: 40 Duration: 2 hours

Answer all questions. Figures in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. Read the following code and detect different types of errors. 8

```
1. include<string.H>
2. int x, y = 2.5;
3. int x_arr(char a[], int n,){
4.     tor(intt i=0; a[i]<n; i++)
5.         scanf("%s",a[0]);
6.     return i;
7. }

8. int main(){
9.     char a1 = "0", a2[10],, x[] = 'abc', 3a;
10.    strcpy(a2,x,5);
11.    switch(5/3){
12.        case 0: printf("Hi!!! This is CASE 0\n");
13.        case 1: print("Hi!!! This is CASEE 1\n ");
14.        printf("%f is a number\n"2.3);
15.        printf("count: %ld", X_arr (a2));
16.        printf("Addition of two number is : %d",5-0)
17.        return 2;
18.}
```

2. Generate a recursive predictive parsing table for the following grammar. [3]

$A \rightarrow 12a \mid [A] \mid 11B$
 $B \rightarrow +C3 \mid - \mid 2B \mid C \mid \epsilon$
 $C \rightarrow 2G \mid +1aD \mid a1C$
 $D \rightarrow a \mid +2D \mid [1]B2a \mid a+2 \mid +B1 \mid \epsilon$

3. Construct a Non-recursive Predictive Parsing table for the following grammar and check whether string **"(11 + 10) * 1"** is accepted or not. [3]

$E \rightarrow E+E \mid E * E \mid (E) \mid N$
 $N \rightarrow 0N \mid 1N \mid 0 \mid 1$

4. Eliminate immediate left recursion from the following grammar. [3+3]
- a) $U \rightarrow eXa \mid aUf \mid UYfZ \mid UY \mid UYZg \mid XZ \mid ZXZ \mid UZZZ$
 $X \rightarrow XXaY \mid XaXgXY \mid XaXX \mid aYZg \mid YaXb \mid ZX \mid ZaY \mid \epsilon$
 $Y \rightarrow OXO \mid OYO \mid YvY10 \mid Y1Y10 \mid YX \mid ZY \mid OYY \mid YYYOY$
- b) $Q \rightarrow R^*Qb \mid P+Rc \mid Q/Rd \mid Q-Pe \mid P^*Q^*R \mid P \mid Q \mid SPQR \mid \epsilon$
 $P \rightarrow P^*Q+R \mid R+Pg \mid PP^*Q/R \mid R/Qf \mid /PQ \mid P+Rc \mid Q/Rd$
 $R \rightarrow (R) \mid R+(P^*Q) \mid R/2 \mid P^*Q+R \mid R+Pg \mid R^*Qb \mid P+Rc \mid R$
5. Left Factor the following Variables. [4+4]
- a. $X \rightarrow \text{Rabit Laughs} \mid \text{Rabit Loves Radish} \mid \text{Rabits talks} \mid \text{Rabit Loving} \mid \text{refrening} \mid \text{restrained} \mid \text{reference} \mid \text{Rabits} \mid \text{Rabit Lashes} \mid \text{refer} \mid \text{Rabit Lost} \mid \text{Rabit Loving} \mid \text{Rabit Lasting} \mid \text{Rabit} \mid \text{referee} \mid \text{ref}$
- b. $Y \rightarrow [23] \mid [2234] \mid 345 \mid 33iop \mid 33kip \mid 33iol \mid 33jle \mid 32ir \mid 322pl \mid 324 \mid 322yu \mid 345uit \mid 345uk \mid 33ilp \mid 33ikol \mid 33kkpo \mid 33ki \mid 33k \mid [1yp] \mid [2yui] \mid 36hy \mid 4$
6. For the following grammars, write down the FIRST and FOLLOW sets for each nonterminal. [4+4+4]
- (a) $X \rightarrow eXYT-h \mid iXPrq \mid YtPrq \mid Pgh \mid cd$
 $Y \rightarrow pqX^*P \mid c \mid d \mid Piy \mid TroXg \mid \epsilon$
 $P \rightarrow Yq \mid PcTXp \mid TiqXfYqP/ \mid cP \mid \epsilon$
 $T \rightarrow x \mid y \mid Tp \mid PiXr \mid \epsilon$
- (c) $P \rightarrow ^2P3Q \mid QyPr \mid PRok \mid \epsilon$
 $Q \rightarrow R3P\% \mid TTxR \mid P3Q \mid b$
 $R \rightarrow uRa \mid RRjQy \mid c \mid \epsilon$
 $T \rightarrow xTd \mid RR \mid g \mid ThPr \mid TdQQ$
- (b) $A \rightarrow uAB \mid vBg \mid Agh$
 $B \rightarrow xC+A) \mid yCrAt \mid b \mid Big \mid \epsilon$
 $C \rightarrow DaC \mid +23 \mid A1 \mid D$
 $D \rightarrow *2 \mid \%3 \mid aDr \mid D\&2 \mid \epsilon$